

3.3.2 Alkanes

Alkanes are the main constituent of crude oil, which is an important raw material for the chemical industry. Alkanes are also used as fuels and the environmental consequences of this use are considered in this section.

3.3.2.1 Fractional distillation of crude oil

Content	Opportunities for skills development
Alkanes are saturated hydrocarbons. Petroleum is a mixture consisting mainly of alkane hydrocarbons that can be separated by fractional distillation.	AT a, d and k PS 1.2 Fractional distillation of a crude oil substitute.

3.3.2.2 Modification of alkanes by cracking

Content	Opportunities for skills development
Cracking involves breaking C–C bonds in alkanes. Thermal cracking takes place at high pressure and high temperature and produces a high percentage of alkenes (mechanism not required). Catalytic cracking takes place at a slight pressure, high temperature and in the presence of a zeolite catalyst and is used mainly to produce motor fuels and aromatic hydrocarbons (mechanism not required). Students should be able to: explain the economic reasons for cracking alkanes.	

3.3.2.3 Combustion of alkanes

Content	Opportunities for skills development
Alkanes are used as fuels. Combustion of alkanes and other organic compounds can be complete or incomplete. The internal combustion engine produces a number of pollutants including NO_x, CO, carbon and unburned hydrocarbons. These gaseous pollutants from internal combustion engines can be removed using catalytic converters. Combustion of hydrocarbons containing sulfur leads to sulfur dioxide that causes air pollution. Students should be able to: • explain why sulfur dioxide can be removed from flue gases using calcium oxide or calcium carbonate.	

3.3.2.4 Chlorination of alkanes

Content	Opportunities for skills development
The reaction of methane with chlorine. Students should be able to: • explain this reaction as a free-radical substitution mechanism involving initiation, propagation and termination steps.	